



JDBC CRUD Operations and API Overview

Section 1: Learn

What is API?

API stands for **Application Programming Interface**. It is a set of classes, interfaces, and methods that allow different software components to communicate.

In JDBC, the API enables Java programs to interact with databases using SQL.

Interfaces of JDBC API

Interface	Description
Connection	Establishes a connection to the database
Statement	Used to execute static SQL statements
PreparedStatement	Used for precompiled and parameterized SQL
ResultSet	Represents the result of a SQL query

Classes of JDBC API

Class	Description
DriverManager	Manages a list of database drivers
SQLException	Handles SQL-related



Class	Description
ption	errors

Section 2: Practice

Creating Data – INSERT

Using **PreparedStatement** (more secure than **Statement**):

```
String query = "INSERT INTO students (name, age) VALUES (?, ?)";
PreparedStatement pstmt = conn.prepareStatement(query);
pstmt.setString(1, "Ravi");
pstmt.setInt(2, 22);
int rows = pstmt.executeUpdate();
System.out.println(rows + " row(s) inserted.");
```

Reading Data – SELECT with ResultSet

```
String query = "SELECT * FROM students";
Statement stmt = conn.createStatement();
ResultSet rs = stmt.executeQuery(query);

while (rs.next()) {
    System.out.println(rs.getInt("id") + " " + rs.getString("name"));
}
```



Updating Data – UPDATE

```
String query = "UPDATE students SET age = ? WHERE id = ?";
PreparedStatement pstmt = conn.prepareStatement(query);
pstmt.setInt(1, 23);
pstmt.setInt(2, 1);
int rows = pstmt.executeUpdate();
System.out.println(rows + " row(s) updated.");
```

Deleting Data – DELETE

```
String query = "DELETE FROM students WHERE id = ?";
PreparedStatement pstmt = conn.prepareStatement(query);
pstmt.setInt(1, 1);
int rows = pstmt.executeUpdate();
System.out.println(rows + " row(s) deleted.");
```

Section 3: Know More (FAQs)

Why use PreparedStatement over Statement?

- Prevents **SQL Injection**
- More **efficient** for repeated executions

Can I execute multiple queries with one statement?

Yes, using `addBatch()` and `executeBatch()`.

What is the difference between `executeQuery()` and `executeUpdate()`?

- `executeQuery()` is for **SELECT**, returns **ResultSet**



- `executeUpdate()` is for `INSERT`, `UPDATE`, `DELETE`, returns rows affected

Do I need to close ResultSet and Statement?

Yes. Always close `ResultSet`, `Statement`, and `Connection` to free resources.

What if I forget to close the connection?

It may lead to **memory leaks** and **connection limit issues** in the database.